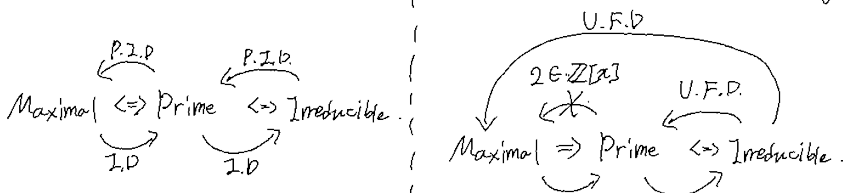
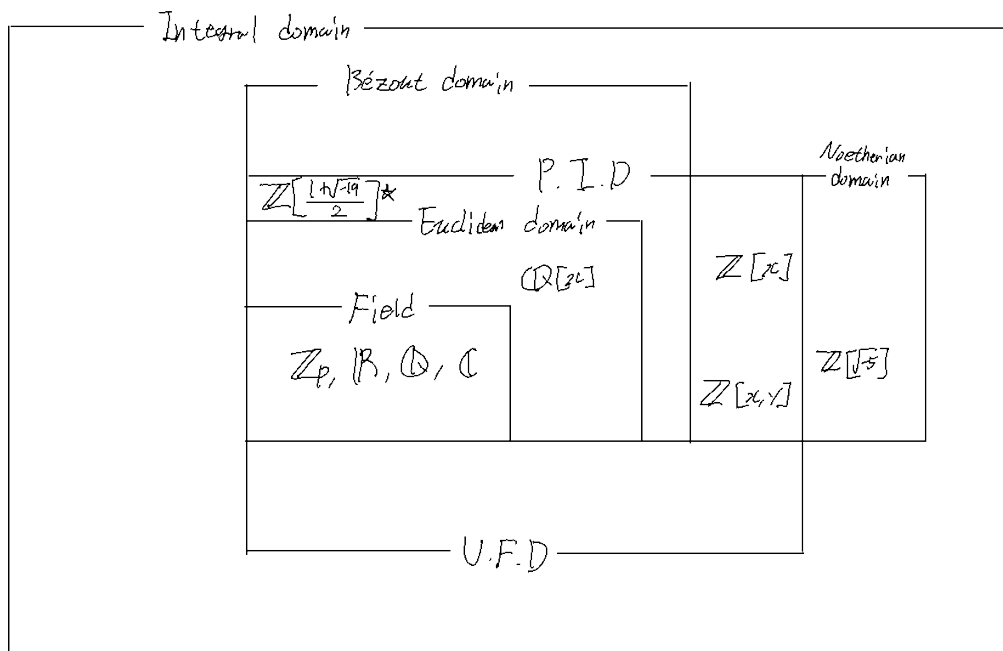


Field $\Rightarrow$ Euclidean Domain $\Rightarrow$ Principal Ideal Domain $\Rightarrow$ Unique Factorization Domain $\subset$ Integral Domain.



P.I.D. $\Leftrightarrow$ Noether $\Leftrightarrow$ Every ideal are finitely generated $\Leftrightarrow$ ACC $\Rightarrow$ A.C.C.P $\Rightarrow$ Factorization $\Leftrightarrow$ U.F.D.

Bézout $\Leftrightarrow$ Finitely generated ideal is principal $\Rightarrow$ irreducible = prime $\Rightarrow$ Factorization $\Leftrightarrow$ U.F.D.



Basic languages for generated ideal.

Def. Let R be a Commutative Ring, and $a, b \in R$ with $b \neq 0$.

1. a is called a multiple of b if: there exists $x \in R$ s.t. $a = bx$. Simply, $b|a$.
2. A non-zero $d \in R$ is called a Greatest Common divisor of a and b if:

$$d|a \text{ and } d|b \text{ (or, } a, b \in (d)), \quad d'|a \text{ and } d'|b \Rightarrow d'|d.$$
$$(a, b \in (d') \Rightarrow (d) \subseteq (d'))$$

Note: $b|a \Leftrightarrow a \in (b) \Leftrightarrow (a) \subseteq (b)$.

Directly, $(a, b) = (d)$ implies d is the g.c.d of a and b ,
but conversely, gcd maybe does not generate the ideal in $\mathbb{Z}[x]$,
 $(1) \neq (2, x)$ but 1 is gcd of 2 and x

Prop. Let R be a Commutative Ring, and $a, b, c \in R$ be given. Then,

- 1). $(a, b) = (a) + (b)$
- 2). $(c) \cdot [(a) + (b)] = (c) \cdot (a) + (c) \cdot (b) = (ca) + (cb)$.
- 3).

